

PARANA, Argentina

WMO: 873740

Lat: 31.7903S

Lon: 60.4842W

Elev: 256

StdP: 14.56

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.9	39.2	25.5	19.7	45.9	28.4	22.6	46.9	25.9	55.2	23.0	56.5	3.4	180	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.6	94.0	73.7	91.5	73.0	89.3	72.3	78.2	88.5	76.6	86.5	75.2	84.7	9.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	133.8	84.5	73.7	126.6	82.5	72.2	120.3	80.8	42.0	88.6	40.4	86.6	38.9	84.8	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	20.8	18.4	32.3	98.8	2.3	2.0	30.6	100.2	29.2	101.4	27.9	102.5	26.2	104.0		
(5)			29.9	81.2	2.5	1.8	28.1	82.5	26.6	83.5	25.2	84.5	23.4	85.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.6	77.6	75.2	71.4	65.7	59.7	55.0	54.0	56.8	60.5	65.8	70.7	74.9		
	DBStd	10.14	4.91	5.13	5.42	6.33	6.38	6.72	7.34	7.70	7.00	6.12	5.63	5.39		
	HDD50	85	0	0	0	0	3	22	38	19	3	0	0	0		
	HDD65	1459	0	1	11	67	190	311	354	277	169	62	15	2		
	CDD50	5765	855	706	663	470	305	172	161	230	320	491	621	771		
	CDD65	1663	391	287	208	86	27	11	11	24	35	89	186	308		
	CDH74	14783	4060	2532	1597	518	96	22	40	169	317	725	1671	3036		
	CDH80	5541	1814	992	503	105	12	2	4	35	83	189	558	1244		
(14) Wind	WSAvg	7.4	6.8	6.7	6.2	6.5	6.4	7.1	7.5	8.1	8.5	8.9	8.1	7.7		
(15) Precipitation	PrecAvg	42.2	5.2	4.8	5.9	4.7	2.5	1.4	1.3	1.5	2.3	4.4	4.6	5.0		
	PrecMax	60.9	38.7	15.1	21.2	15.5	9.1	4.6	3.4	6.7	9.7	12.9	12.3	15.9		
	PrecMin	20.8	0.4	0.2	0.9	0.2	0.0	0.0	0.0	0.0	0.1	0.6	0.3	0.6		
	PrecStd	9.3	5.6	3.4	3.7	3.7	2.2	1.2	0.9	1.5	2.0	2.7	2.6	3.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.0	95.3	91.7	87.0	81.5	76.9	79.0	84.5	87.7	90.0	92.1	96.2		
		MCWB	75.8	74.9	74.4	74.4	70.8	69.3	66.4	69.1	71.5	72.6	71.1	73.1		
	2%	DB	94.5	91.7	88.4	83.0	76.7	72.3	73.6	78.7	82.0	84.7	88.8	92.7		
		MCWB	73.7	74.4	73.2	71.1	67.8	66.5	64.8	65.8	66.8	68.8	69.8	72.7		
	5%	DB	91.6	89.1	85.5	79.8	73.1	68.5	69.4	74.3	77.4	81.1	86.1	89.7		
		MCWB	72.9	73.5	71.9	69.5	66.2	63.2	62.0	63.4	64.2	67.4	68.5	71.7		
	10%	DB	89.2	86.3	82.5	76.6	69.8	65.4	65.6	69.9	73.3	77.8	83.0	86.9		
		MCWB	72.4	72.4	70.1	67.7	63.6	60.2	59.6	60.5	61.0	65.1	67.4	70.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.1	80.0	78.5	76.5	72.4	70.4	68.5	70.9	73.4	76.0	75.9	79.7		
		MCDB	90.3	88.5	88.4	84.1	79.1	75.2	75.5	81.6	84.2	86.8	86.4	90.7		
	2%	WB	77.7	77.8	76.2	73.9	69.4	67.4	65.5	67.5	69.4	72.2	72.8	77.0		
		MCDB	89.0	87.2	84.1	80.4	74.0	71.3	71.9	76.4	79.4	80.4	83.8	87.0		
	5%	WB	76.2	76.0	73.9	71.5	67.3	64.7	63.2	65.2	66.1	69.7	71.0	74.6		
		MCDB	87.2	84.8	81.4	77.1	71.9	68.3	68.4	72.6	73.9	77.8	81.0	84.7		
	10%	WB	74.6	74.2	71.8	69.3	65.1	61.9	60.9	62.0	63.3	67.4	69.3	72.5		
		MCDB	85.0	82.7	79.8	74.4	69.2	64.5	64.9	68.2	70.8	75.0	79.0	82.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.6	18.0	17.7	16.1	14.8	14.6	15.5	17.7	18.7	18.4	19.9	19.7		
		MCDBR	22.6	21.5	21.1	19.8	17.5	16.1	18.0	22.0	24.1	22.9	23.8	22.9		
	5% WB	MCWBR	8.7	8.7	9.2	10.0	9.8	10.1	10.4	12.3	12.0	11.1	10.4	9.5		
		MCWBR	19.2	17.9	18.0	16.5	14.7	14.0	15.6	19.7	20.6	19.5	19.8	19.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.400	0.388	0.372	0.380	0.358	0.355	0.358	0.425	0.485	0.426	0.392	0.396		
		taud	2.384	2.433	2.468	2.413	2.436	2.435	2.397	2.159	1.997	2.240	2.363	2.378		
	Ebn at Noon	Ebn at Noon	298	295	289	268	258	250	255	248	249	280	298	300		
		Edh at Noon	41	38	34	32	28	27	29	41	54	45	42	41		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2266	1945	1675	1233	915	790	879	1163	1512	1795	2187	2292		
	RadStd	RadStd	148	149	139	157	80	86	93	104	131	150	126	135		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (3 neighbors)	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
		N/A	N/A	N/A	N/A	+0.90	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	+0.41	+0.88	+0.99	N/A	N/A	N/A

Nomenclature: See separate page